# UNIT 13

# SPECTROSCOPY

## SPECTROSCOPY:

"The interaction of EMR to the matter is simply a modern analytical technique known as Spectroscopy."

## UV-VIS (ULTRAVIOLET-VISIBLE) SPECTROSCOPY:

UV-VIS (ultraviolet-visible) spectroscopy or spectrophotometry is the study of the interaction of light with matter at electronic levels.

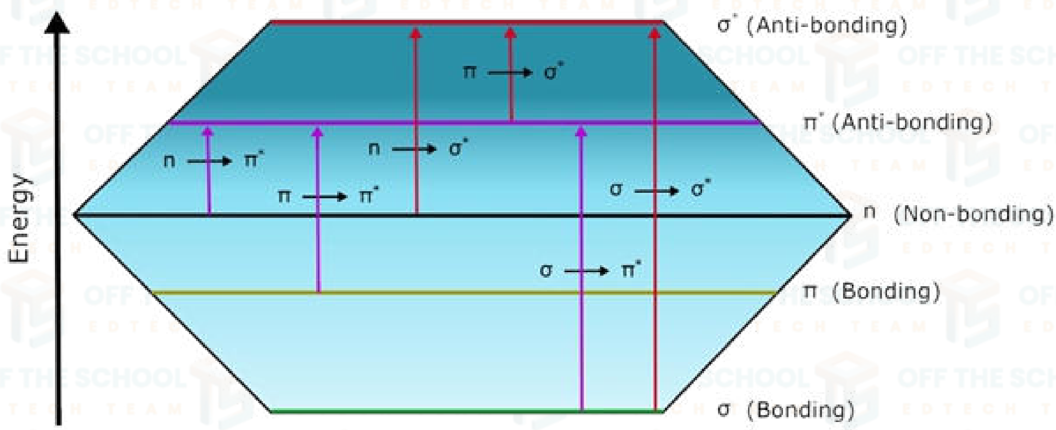
• It ranges from ultraviolet region i.e. 200nm to the visible region i.e. 800nm.

• UV spectrum extends from 200nm to 400nm whereas the visible region ranges from 400nm to 800nm.

## PRINCIPLE:

• Energy absorbed in the UV-Visible region produces changes in the electronic energy of the molecule resulting from transition of valence electrons in the molecule.

• The distinct types of electrons are involved in organic molecules. These are as follows: σ, π & n Electrons.




**Transition in Molecular Orbitals**

**Energy of σ-σ* > n-σ*> π-π*> n- π***

• **σ Electrons:** These electrons are involved in saturated bonds, and known as σ bonds. These bonds are not excited by UV-visible radiation.

• **π Electrons:** These electrons are involved in unsaturated hydrocarbons.

• **n-Electrons:** These are the electrons present when hetero atoms are present like nitrogen, oxygen, halogens etc.

• Any molecule has either n, π, σ or a combination of these electrons; these bonding (π or σ) and non-bonding (n) electron absorbs the characteristic radiation and undergoes transition from ground state to excited state.

## TRANSITIONS:

**(i) σ-σ* Transition:** High energy required for this transition | Falls in the vacuum UV region (150nm).

Occurs in saturated compounds like hydrocarbons (methane, ethane etc).

**(ii) n-σ* Transition:** Lesser energy than σ- σ*. Falls in the near UV region (150-300nm).

Occurs in saturated compounds with heteroatoms like Alcohols, Water etc.

E.g. $CH_2OH$, $C_2H$,OH, $H_2O$, $CH_2I$, CH3-S-CH3 (heteroatoms like O/N/S/Halogens).

**(iii) π-π* Transition:** Less energy required. Falls in the range of 180-320nm based on conjugation.

Unsaturated compounds having double/triple compounds, Alkenes, Alkynes. Occurs in Aromatic compounds.

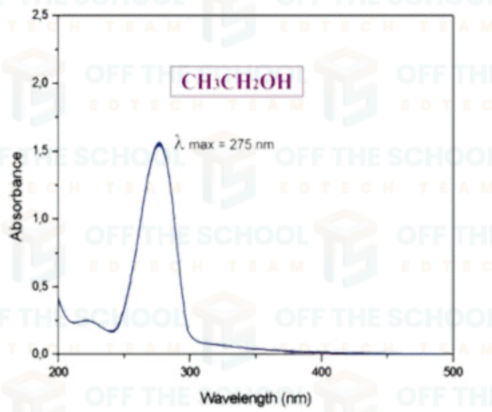
**(iv) n–π* Transition:** Lowest energy is required. The range is 200-800nm. Occurs in unsaturated having heteroatoms like carbonyl and ketones etc.

**Explain λmax with the help of U.V-visible spectrum of ethanol?**

Ethanol absorbs light in the UV-Visible region due to an n ⟶ π* transition in the hydroxyl group. This transition occurs when an electron in the non-bonding orbital (n) of the oxygen atom is excited to an anti-bonding π* orbital.

The λmax of ethanol is 275 nm, which corresponds to the energy required to excite an electron from the n orbital to the π* orbital.

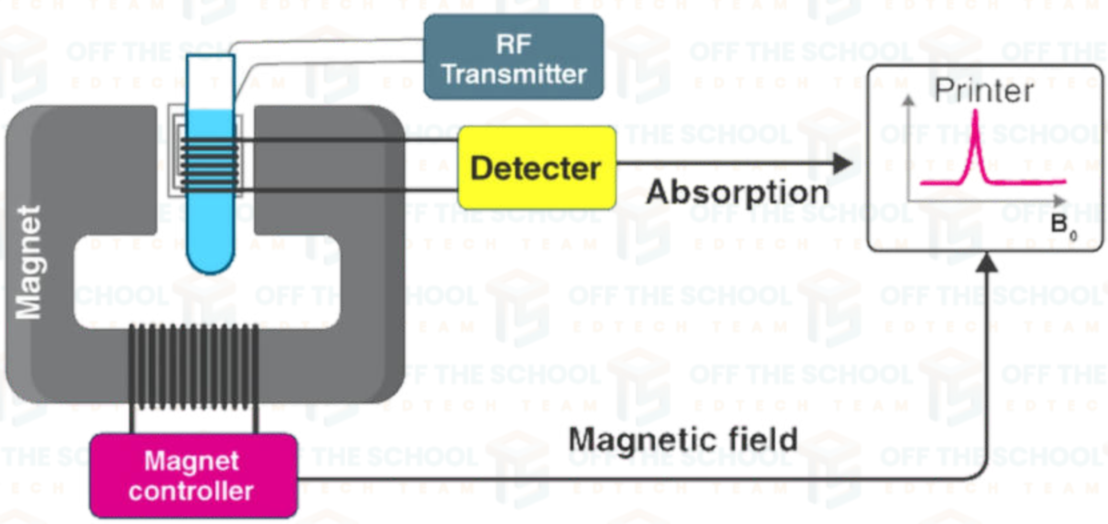
The absorption of light by ethanol can be used to identify it in a mixture of other compounds, or to determine its concentration in a solution.




U.V-visible spectrogram of pure ethanol ($C_2H_5OH$)

**NMR (NUCLEAR MAGNETIC RESONANCE) SPECTROSCOPY:**

**Definition:** (NMR) spectroscopy is the study of molecules by recording the interaction of radiofrequency (Rf) with the nuclei of molecules placed in a strong magnetic field.




**NMR spectroscopy Instrumentation**

**Basic principle:**

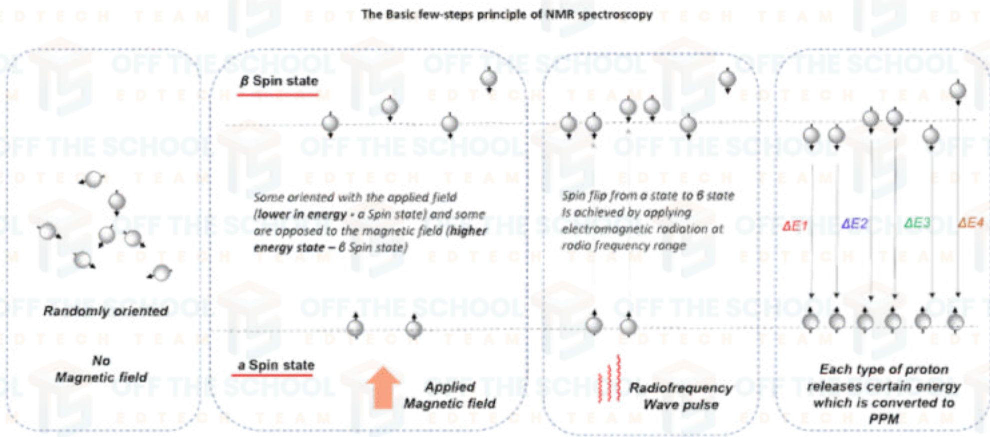
Here is the basic principle of NMR spectroscopy:

**1.** We put the sample in a magnetic field.

**2.** The nuclei distribute in two different energy levels – some oriented with th applied field (lower in energy – α Spin state) and some are opposed to th magnetic field (higher energy state – β Spin state)

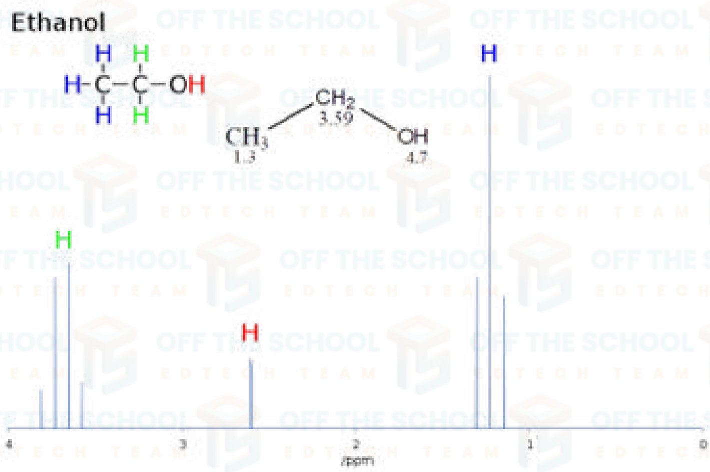
**3.** Electromagnetic radiation matching this energy difference (radio frequency) i applied.

**4.** Each proton absorbs energy exactly matching the gap (resonance) betwee the α and β states causing a spin-flip.




**5.** Protons relax back each releasing energy which is converted to the δ (ppm) value on the spectrum.

**NMR spectrum:**
In NMR spectrum, X-axis represents the chemical shift of proton signals relative to TMS (tetramethylsilane or trimethylsilane).
Most of the protons has chemical shift between 0-12ppm while TMS has standard value of 0 ppm.
Y-axis represents absorption shows intensity of NMR signals.
Peaks represents the splitting pattern (singlet, doublet, triplet, and quartet) due to neighboring proton.




Example 1H NMR spectrum (1-dimensional) of ethanol plotted as signal intensity vs. chemical shift. There are three different types of H atoms in ethanol regarding NMR. The hydrogen (H) on the -OH group is not coupling with the other H atoms and appears as a singlet, but the CH3- and the -CH2- hydrogens are coupling with each other, resulting in a triplet and quartet respectively.

## Atomic Absorption Spectroscopy (AAZS):

**1. Principle:**
• **Absorption:** AAS measures the amount of light absorbed by atoms in a sample.
**2. Light Source:**
• **Continuous Spectrum:** Uses a continuous spectrum light source (like a hollow cathode lamp).
**3. Analyte Detection:**
• **Presence of Atoms:** Detects the presence of specific atoms in the sample.
**4. Process:**
• **Absorption of Light:** Atoms in the sample absorb specific wavelengths of light, and the amount of absorption is used to quantify the concentration of the element.
**5. Energy Transition:**
• **Ground to Excited State:** Atoms absorb energy and transition from the ground

*Absorbtion of Hydrogen atom*

**Atomic Emission Spectroscopy (AES):**

**1. Principle:**

• **Emission:** AES measures the light emitted by atoms in a sample.

**2. Light Source:**

• **Discrete Spectrum:** Uses a discrete line spectrum light source (like a flame or plasma).

**3. Analyte Detection:**

• **Emission of Atoms:** Detects the emission lines of specific atoms in the sample.

**4. Process:**

• **Emission of Light:** Atoms in the sample emit light at specific wavelengths when returning from an excited state to the ground state.

**5. Energy Transition:**

• **Excited to Ground State:** Atoms release energy as light when transitioning from an excited state to the ground state.

*Emission of Hydrogen atom*